Measuring the Overhead of Trust Mechanisms in the X-Road Data Exchange Layer

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***Abstract*—X-Road secures data exchange between organisations with signatures, certificate status checks, time-stamps and a message log, and proposals to change these mechanisms need a baseline for what they cost now. This study measures the overhead of the trust mechanisms of X-Road 7.8.3 per message on a testbed of three laptops connected by a switch, on one of them alone, on a faster laptop, and with a controlled round-trip time added between the Security Servers. Since the mechanisms cannot all be switched off, each is observed by changing its configuration, through per-stage timings and through packet captures. With one client the X-Road path adds about 28 ms to a 10 KB exchange. Writing the body to the message log accounts for almost half of the latency of a 1 MB exchange, and synchronous time-stamping adds about 90 ms and lowers the highest throughput to 37–47 %, while status checks and batched time-stamps stay out of the request path. With an added round-trip time the default policy of opening a new connection for every message has the largest effect among the configurations compared: each message waits for about five round trips at 10 KB and ten at 1 MB, against two with reused connections, which saves 54 % and 72 % of the latency at 50 ms. The cost of distance between Security Servers thus depends strongly on the connection policy.**

I. Introduction

When one organisation acts on data received from another, it depends on something it did not produce and cannot check by itself. Trust is the name for that dependence. In organisational research it is defined as the willingness of a party to be vulnerable to the actions of another party, based on the expectation that the other will perform an action important to the trustor, without being able to monitor or control it [1]. In computing the same idea is phrased as a belief in the competence of an entity to act dependably, securely and reliably within a specified context [2]. In security engineering a *trusted* component is distinguished from a *trustworthy* one: the first is a component whose failure can break the security policy, the second one that will not fail [3]. A well-designed system therefore tries to need as little trust as possible, and to place the rest in a few components that are named explicitly and protected accordingly.

X-Road, the data exchange layer, is built on this principle. Its member organisations do not have to trust each other's systems directly. Their trust rests on a root of trust, a point that is accepted as trusted axiomatically [4]. In X-Road this root is centralised: it is formed by the certification authorities that vouch for the members' keys [4], together with the Central Server, which publishes who the members are and which authorities to trust, and the time-stamping authority [5]. Every message then carries evidence that can be checked against this root. It is signed by the sender's Security Server and verified by the receiver's, written into the message logs on both sides and time-stamped, and it travels over a mutually authenticated Transport Layer Security (TLS) channel, while the status of the certificates is checked through the Online Certificate Status Protocol (OCSP) [5], [6]. Trust in the other party is replaced by verification against the root of trust, and an exchange can later be used as evidence of who sent what and when. The other side of a centralised root is that it becomes a single point of failure [4], which is why alternatives to it are being studied.

Verification is not free. Signing, verification, logging and setting up the channel take computation and bytes, some of them on every message and some in the background. X-Road has been measured before, but end to end, for questions other than the cost of these mechanisms: a study of threshold signatures reports end-to-end round trips of about 32 ms with all components on one laptop and presents them as a lower bound [7]; a capacity study measures the overhead of a Security Server in front of one registry service, where database time dominates [8]; and the paper that introduced batch time-stamping gives an analytical estimate and leaves benchmarking of the implementation to future work [6]. No performance data published by the Nordic Institute for Interoperability Solutions (NIIS), which develops X-Road, was found. None of these sources splits the cost of an exchange by mechanism.

Two developments make that missing split more relevant. The first is volume: the design of batch time-stamping was calibrated against about 240 million queries per year [6], while the Estonian instance now reports 3.74 billion [9], so whatever is paid per message is multiplied by a much larger base. The second is a set of proposals that would make the trust mechanisms themselves more expensive. Post-quantum signatures are expected to enlarge certificates, signatures and handshakes [10], [11]; a Dilithium5 signature, for example, takes 4595 bytes against the 256 bytes of RSA-2048 [11]. Decentralised Public Key Infrastructure (PKI) built on verifiable credentials adds a presentation to every request [12], and threshold signatures replace a local key with a distributed protocol [7]. These works describe what will become more expensive, but without a measured baseline for the present system they cannot say by how much.

The proposals above do not change X-Road as a whole but individual mechanisms. Post-quantum algorithms affect

signing, verification and the TLS handshake [10], a decentralised PKI changes how certificates are checked [12], and threshold signatures change how messages are signed [7]. The cost of each change can only be judged against the cost of the mechanism it replaces, and since X-Road keeps certificate status checking and time-stamping out of the request path by design [6], it also matters which of these costs are paid on every message. Larger post-quantum signatures and certificates make messages bigger, and the growth in volume means more messages processed at the same time, so the costs have to be known as a function of message size and load. Finally, the most recent figures were obtained with all components on one machine, which their authors present as a lower bound [7], whereas the Security Servers of two organisations normally communicate over a network, and the NIIS report expects post-quantum handshakes to add round trips on it [10]. This leads to the following research questions:

RQ1 How much do the trust mechanisms of an exchange contribute to its latency and to the bytes between the Security Servers, as far as they can be separated from each other, and which of them are paid on every message rather than deferred?

RQ2 How do those contributions change with message size, and how does the throughput of each configuration change with the number of concurrent clients?

RQ3 Does the answer change when the Security Servers run on separate machines instead of one host, and how does it change with a round-trip time added between them?

The contribution is a measurement rather than a new method: a testbed that can be redeployed on one or several hosts, the differences between configurations in milliseconds and bytes and the number of background requests from three environments, a descriptive per-stage breakdown taken from X-Road's own operational monitoring, the effect of a controlled added round-trip time together with a prediction made before that measurement and refined with packet captures, and the per-run data with their summaries. The study measures X-Road as it runs today, with its default keys: post-quantum schemes are not yet available in X-Road [10], [11], so their effect is discussed rather than measured.

II. Background: what is paid per message

X-Road is a centrally managed distributed data exchange layer. A Central Server distributes the global configuration — the member registry, the trusted Certification Authorities (CAs) and the trusted time-stamping authorities — and each member operates a Security Server that handles key management, signing, secure transport, logging and access control [12], [5].

For this study the mechanisms divide by their position relative to the request path:

- **Synchronous**, paid on every message: the mutually authenticated TLS channel between the Security Servers; signing with the member's signing key; verification of the incoming signature; and writing request and response into the message log, whose bodies are logged in full by default [13]. How much the TLS channel costs per message depends on whether connections are reused [14]. By default they are not: the client-side pool reuse (`pool-enable-connection-reuse`) and the provider-side support for pooled connections (`server-support-clients-pooled-connections`) are both off, so each message opens a new TCP connection and performs a TLS handshake [13].
- **Asynchronous**, amortised or deferred: certificate status checking, which uses cached OCSP responses refreshed on a fixed interval (20 minutes by default) rather than one query per message; time-stamping, applied in batches over a hash chain of log records (once per minute by default, with a tolerated time-stamping outage of four hours, the parameter `acceptable-timestamp-failure-period` being 14400 seconds); and the periodic refresh of the global configuration [15], [13], [6]. Because one time stamp covers a whole batch, its cost per message is divided by the size of the batch and falls as the load grows [5], [6].

The split is a documented design decision: caching status responses and deferring time stamps were introduced to keep a per-message PKI operation from becoming both a single point of failure and a latency source [6]. It carries a security price as well, since a cached, nonce-less OCSP response leaves a window during which a revoked certificate is still accepted [12]. The performance side of that trade-off is the subject of this study.

III. Methodology

A. Testbed

The testbed runs X-Road 7.8.3 in containers built from the official X-Road images: one Central Server, a test CA that also provides the OCSP and time-stamping services, and four Security Servers — a management server, one consumer (*ss1*) and two providers (*ss2*, *ss3*). The measured path is *load generator* → *ss1* → *ss2* → *echo service*; the second provider is used only to check what limits the consumer under load. The echo service answers a REST request with a body of the requested size, filled with random bytes so that the message log cannot compress it, and its own processing time is negligible. The load generator is k6, sending REST `GET` requests in a closed loop with a given number of concurrent clients, so the message size below is the size of the response body. The time-stamping service of the test CA was replaced by a multi-threaded implementation, because the original one handles one request at a time and stalled under synchronous time-stamping.

Every Security Server is limited to 2 CPU cores and 4 GB of memory and the Central Server to 2 GB, in all environments, so that the environments differ in where the containers run rather than in how much hardware each of them receives. The whole testbed, including member and service registration, is

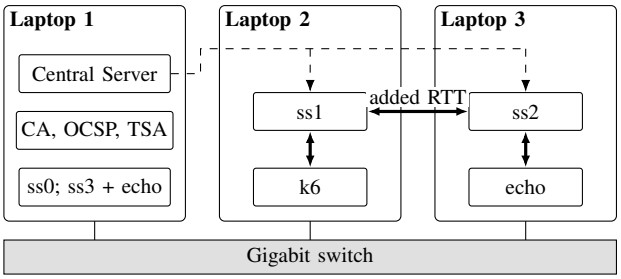


Figure 1. The testbed on three machines, with the test CA providing OCSP and time-stamping (TSA). Thick line: the measured path. Dashed line: background traffic to the central services, which is not delayed. All traffic between the Security Servers passes through the switch.

set up by scripts from a single composition file and can be redeployed from scratch.

B. Environments

a) Three machines.: Three identical laptops (Intel Core i5-8250U, 4 cores and 8 threads, 16 GB of memory) connected by a dedicated wired switch. The first runs the Central Server, the test CA, the management server and the second provider, the second runs the consumer and the load generator, and the third runs the provider and the echo service. The containers have their own addresses on the switch network (Docker `ipvlan`), so the traffic between the Security Servers is not translated by the host and leaves the machine. Clocks are synchronised with chrony to within half a millisecond, which matters when timestamps from two machines are compared. Figure 1 shows the layout.

b) One machine.: The same stack on one of these laptops alone. With the same CPU and the same container limits, the difference to the three-machine environment is the network between the Security Servers and the sharing of one CPU by all components.

c) Development laptop.: The same stack on a newer laptop (Intel Core Ultra 7 265U) under Windows 11, with the containers running in WSL 2. It shows how the absolute numbers depend on the CPU.

d) Emulated network.: In all of the environments above the Security Servers are at most one switch apart, with a round-trip time below a millisecond, whereas the Security Servers of two organisations usually communicate over a longer and less predictable path. From local measurements alone it is not possible to tell whether the results would hold there. A controlled round-trip time is therefore added between the consumer and the provider on the three machines with netem, the network emulator of the Linux kernel [16], which adds a chosen delay to the packets of a given flow. This is not a model of a real wide-area deployment, only of the added round-trip time. The same approach is used to study TLS under different network conditions without Internet-scale infrastructure [17], and the accuracy of the delay netem produces has been examined in [18], [19]. How the delay is applied and verified is described in Section III-F.

Table I
HOW THE COST OF EACH TRUST MECHANISM IS OBSERVED.

Mechanism	Observed through
The whole X-Road path	*direct* compared with *full*
Inter-server connection policy	*reuse* compared with *full*
Logging of message bodies	*nobody* compared with *full*
Signing and verification	stages of the breakdown (Section III-D)
Time-stamping	requests to the time-stamping service; *synctsa* as another policy
Certificate status checking	requests to the OCSP responder

On all machines the CPU runs in its performance power mode, which is checked before every grid and recorded with it, since the power mode affects the latency.

C. Isolating the cost of a mechanism

The trust mechanisms of RQ1 cannot all be switched off one at a time, so each of them is observed in the way that changes as little else as possible (Table I).

Where a setting removes or changes a mechanism, a configuration is compared with the default one (*full*) and differs from it in that setting only:

- **direct** — plain HTTP from the load generator to the echo service, without X-Road. Its difference to *full* is the whole X-Road path, not any single mechanism.
- **reuse** — connections between the Security Servers are kept open (`pool-enable-connection-reuse`, `server-support-clients-pooled-connections` and `server-connector-max-idle-time`). By default each message opens a new connection [13], so the difference is the effect of the connection policy between the Security Servers: setting up the TCP connection and the TLS session and starting every transfer on a new TCP connection [14]. It does not isolate the TLS handshake.
- **nobody** — message bodies are not written to the message log (`message-body-logging=false`), so the difference is the cost of logging the body.
- **synctsa** — every message is time-stamped synchronously (`timestamp-immediately=true`). This is a different policy rather than the cost of the default batched time-stamping, and is reported as such.

Signing and verification cannot be switched off, so they are observed only in the stages of the breakdown in which they happen, together with the other work done in the same stages, rather than as a difference between configurations.

The two background mechanisms cannot be moved into the request path without changing what is measured. Postponing time stamps by a long interval would violate a documented constraint between the time-stamping interval and the OCSP freshness [13]. Making the OCSP freshness shorter than the interval at which the signer refreshes its cached responses makes every request fail, because the Security Server takes

certificate status only from that cache and does not query the responder while it processes a message. For these two mechanisms the cost is therefore observed as the number of requests sent to the time-stamping service and to the OCSP responder during each run and while its messages are being time-stamped.

D. What is measured

For each request the load generator records the latency and whether a correct response of the expected size arrived. For each run the bytes exchanged between consumer and provider, counted by `iptables` inside the network namespace of the consumer on the X-Road port, the growth of the message log, and the number of OCSP and time-stamping requests from the CA log are recorded.

X-Road also records four timestamps per request on each Security Server in its operational monitoring database: when the request arrived, when it was forwarded, when the response arrived and when it was returned. Joining the records of the consumer and the provider by the request identifier splits the latency into six consecutive stages: the consumer before sending, the way to the provider (the stage in which signing and logging on the consumer and setting up the connection take place), the provider before forwarding (the stage in which verification and logging take place), the echo service, the provider returning the response, and the consumer finishing it. The stages add up to the time between the request arriving at the consumer and the response leaving it, which excludes the short way between the load generator and the consumer. The breakdown is descriptive: these are timestamps of the measured system itself with a resolution of 1 ms, and the share of requests whose records could be matched on both servers is reported with it. Stages measured within one host are more reliable than the two stages that span hosts, where the residual clock offset enters the result; explanations that involve the network are checked against packet captures (Section III-F).

E. Procedure

Two message sizes are used with one client. Post-quantum signatures and certificates would add a few kilobytes to every message [11], which matters most for small messages, so a 10 KB response represents a small message; a 1 MB response represents a large one, whose body the message log stores in full by default [13]. One client with no other load gives the baseline latency without queueing, and the load is then increased until one of the components runs out of resources [8]: the number of clients grows beyond the two CPU cores available to each Security Server, to 4, 16 and 32, so as to reach the throughput ceiling of the Security Servers; this range covers the 10 to 30 concurrent clients used in [8]. The main grid therefore crosses the five configurations with 10 KB and 1 MB responses at one client and 10 KB at 4, 16 and 32 clients, plus the two-provider case at 4, 16 and 32 clients.

The Security Server runs on the Java virtual machine, whose first requests after a start are slower until the code is compiled, so only the steady state is measured [20]. After every configuration change the proxy of the Security Servers is restarted and warmed up by a discarded run at four clients, followed by warm-up requests before every measured run. The amount of warm-up is set from a pilot run as the point after which the median latency over consecutive windows of requests stops decreasing. Each run measures 1000 requests, as in [7], and 200 at 1 MB, where every message adds about a megabyte to the message log of each Security Server.

Following [20], every point is measured in several independent runs rather than once. A grid repeats every point five times; within each repetition plain HTTP runs first, then the configurations in random order, each with its points in random order, and a control run with the default configuration at the start of each repetition and at the end shows the drift of the testbed during the grid. Several grids are run and combined. During the measurements the machines run nothing else. Before the message log is cleaned, the runner waits until every logged message has its time stamp, so that deferred work is not lost between runs.

Results are reported as the median over repetitions of each run's median, with the interquartile range between repetitions and the number of successful requests.

F. Added round-trip time between the Security Servers

A controlled round-trip time is added between the consumer and the provider with netem [16], as done for TLS benchmarking in [17]. Half of the round-trip time is added on each side, in the network namespaces of the two Security Servers and only for packets to the other server, so that all traffic between them is delayed, including the OCSP exchange between Security Servers, while the client, the Central Server and the CA stay on the local network. The delay is fixed, without jitter, since netem reproduces jitter less accurately [19], [18]. The results are reported as the effect of a controlled added round-trip time between the Security Servers, not as a measurement of a real wide-area deployment. A control at zero delay with the same netem rules installed checks that the rules themselves do not change the result.

The added round-trip time (0, 5, 20 and 50 ms) is verified on the measured flow itself: from the second delay grid on, the round-trip time of the consumer's connections to the provider is read from the kernel during every run, and a calibration with sequential requests reads it for new connections. Before the first delay grid, traffic captures of sequential requests on the local network show whether a request opens a new connection, with a full or a resumed TLS handshake, or uses an existing one, and how many times one side waits for the other. From these exchanges a prediction is made of how latency should grow with the round-trip time, and it is compared with the measured slope; where they disagree, captures under delay are used to find the waits the prediction missed. Captures of a 1 MB response with and without connection reuse check how the transfer of a large response depends on the state of the TCP connection. Since the delay is applied to outgoing packets, they wait inside the sending host, where they may still count

against the limits of the sender's TCP stack on queued data and so influence how its window grows. As a cross-check, the 1 MB captures and part of the grid are repeated with the delay applied to incoming packets on the receiving side instead.

The delay grid crosses *full* and *reuse* with 10 KB and 1 MB responses at one client and 0, 5, 20 and 50 ms, and an extended grid adds 100 KB responses and 4 and 16 clients at 0, 20 and 50 ms. The cross-check grid repeats *full* and *reuse* with 10 KB and 1 MB responses at one client and 0 and 50 ms.

G. Threats to validity

Several properties of the testbed shift the absolute values. The machines are laptops rather than servers, and the development laptop adds the overhead of WSL 2 on Windows. The time-stamping service of the test CA is a replacement, and the images of the test CA and the load generator were not pinned to a version. The test CA, OCSP responder and time-stamping service run next to the Security Servers, whereas in production they are external services with their own latency, so the figures for status checking and synchronous time-stamping are optimistic. All intervals of the background mechanisms are left at their defaults, while a production deployment may use longer ones: for the Estonian instance of 2017 a master's thesis documents a global configuration validity of six hours and an OCSP freshness of eight hours [21], so background costs per message would be even lower there than in the testbed.

Other properties change the shares and the spread rather than the absolute values. The echo service has no processing time of its own, which raises the relative share of the trust mechanisms compared with a real service. Under load the spread between repetitions is larger, so there the stable throughput and the results of synchronous time-stamping are reported together with the spread and the errors, and small differences between the other configurations are not assigned to individual mechanisms. The added round-trip time is emulated: it has no cross traffic and no losses, and it is applied inside the hosts of the Security Servers; applying it to incoming instead of outgoing packets did not change the slopes (Section IV-D).

Because of these factors the absolute latencies describe this testbed rather than a production deployment. The comparisons are therefore made between configurations on the same machines and under the same conditions, and the results are read primarily as differences between configurations and as their change with message size, load and added round-trip time; absolute values are reported for reference only.

IV. Results

A. Stability of the testbed

The warm-up needed turned out much longer than expected. The closest measurement of X-Road reports means over 1000 requests without describing a warm-up [7], and a first short check on the testbed suggested that about 700 requests would be enough. A longer pilot on the development laptop, with a 1 KB response and before the power mode was fixed (see below), showed otherwise: the latency did not settle within the first thousand requests after a restart of the Security Servers, and the first run after a restart remained slower even after 4000 requests at one client. Repeated runs at the same point differed by up to 20 %. These observations set the warm-up and the number of repetitions described in Section III-E. The effect of the power mode of the CPU was found in the first measurements on three machines. The default configuration took 78 ms with one client, against 41 ms on one of the same laptops alone, and the difference looked like a cost of the network. Changing how the Security Servers acknowledge TCP segments and how the load generator reuses its connection did not change it. The per-stage timestamps then showed all stages, including purely computational ones, 2 to 2.5 times slower than on the development laptop (for example, signing the response on the provider took 22 instead of 10 ms), so the computation had slowed down and the difference was not only a cost of the network. The laptops were in an energy-saving mode, in which an almost idle CPU does not raise its frequency for short bursts of work; on one of them the control run took 79.7 ms in that mode and 28.7 ms in the performance mode. All results below were obtained in the performance mode; the earlier runs in the energy-saving mode are not used.

With five repetitions per grid, the main grid was repeated 30 times on three machines (six grids), 15 times on one machine (three grids) and 20 times on the development laptop (four grids). The control run (*full*, 10 KB response, one client) had a median of 28.7 ms over 36 runs on three machines (27.5 to 37.2 ms, one outlier), 30.4 ms over 18 runs on one machine (28.9 to 31.2 ms) and 26.7 ms over 24 runs on the development laptop (24.3 to 27.9 ms). Every request succeeded except in *reuse* under load, where single requests failed with HTTP 500 in some runs: 7 of 90 000 requests at 4 to 32 clients on three machines, 3 of 45 000 on one machine and 2 of 60 000 on the development laptop. The failures are consistent with a pooled connection that was already closed by the other side.

B. Cost of the mechanisms (RQ1)

Table II shows the latency with one client. The difference between *direct* and *full* is the whole X-Road path: on three machines 27.6 ms for a 10 KB response and 64 ms for a 1 MB response. Logging the message body makes no difference beyond the spread at 10 KB, but at 1 MB not logging it saves 34.9 ms, 47 % of the latency. Keeping connections open between the Security Servers saves 2.0 ms at 10 KB and 4.3 ms at 1 MB. Synchronous time-stamping adds 89–92 ms. The interquartile range between repetitions is 0.7–4.8 % of the median on three machines, up to 9.4 % on the development laptop, and up to 5.1 % on one machine except for *synctsa* (18–31 %).

On three machines a 10 KB exchange carries 14.8 KB from consumer to provider and 25.7 KB back; with *reuse* 11.5 and 22.8 KB, so the connection policy costs 3.3 and 2.9 KB per message in the two directions. The message log of the consumer grows by 33.7 KB per exchange with bodies and 19.3 KB without them, and at 1 MB by 1.42 MB and 19.5 KB.

Table II
MEDIAN LATENCY WITH ONE CLIENT (MS) BY RESPONSE SIZE AND ENVIRONMENT; DIFFERENCES TO *full* IN THE SAME ENVIRONMENT. EVERY REQUEST SUCCEEDED.

	Three machines		One machine		Development laptop	
	10 KB	1 MB	10 KB	1 MB	10 KB	1 MB
direct	0.7	10.0	0.1	0.7	0.1	0.7
full	28.3	74.0	29.8	74.7	24.8	49.7
nobody − *full*	−0.9	−34.9	−0.3	−32.0	−0.9	−17.0
reuse − *full*	−2.0	−4.3	−2.0	−4.3	−1.6	−2.9
synctsa − *full*	+88.9	+91.7	+91.8	+98.1	+62.4	+68.0
repetitions	30		15		20	

The two background mechanisms did not appear in the request path. The median number of requests to the OCSP responder per run was zero; in about 5 % of the runs (35 of 741 on three machines) a background refresh of 2 to 8 requests fell into the run. A configuration with an OCSP freshness shorter than the refresh of the cached responses made every request fail, as described in Section III-C; this was observed in an early run on the development laptop, but it does not depend on the power mode. Batched time-stamping sent a median of zero requests to the time-stamping service during a run (at most two) and about two per run, of about 1200 messages, while the runner waited for the time stamps; *synctsa* sent 4802 per run, four for each exchange, one for every record written to the message log.

Table III shows the per-stage breakdown on three machines. All requests of these runs were matched between consumer and provider. The stage before the consumer sends the request takes 0.1–0.2 ms, so the consumer records the request as forwarded when it starts sending, and signing and logging of the request fall into the stage on the way to the provider. The configurations change the stages in which their mechanism works: at 10 KB *reuse* changes only the way to the provider (−2.4 ms); not logging a 1 MB body shortens the provider's response by 16.9 ms and the consumer's finishing stage by 17.0 ms, that is, logging the body costs about 17 ms on each side; and synchronous time-stamping adds 20–25 ms to each of the four stages in which a record is logged. The two stages that span machines include the residual clock offset of below half a millisecond.

C. Response size and load (RQ2)

From a 10 KB to a 1 MB response, the latency of *full* on three machines grows from 28.3 to 74.0 ms, while *direct* grows from 0.7 to 10.0 ms. Without body logging the growth is from 27.4 to 39.1 ms, so most of the additional time at 1 MB is writing the body to the message log. The effect of the connection policy grows from 2.0 to 4.3 ms, and the cost of synchronous time-stamping stays at about 90 ms.

Under load the spread between repetitions grows to 9–41 % of the throughput for *full*, *nobody*, *reuse* and the two-provider case across the environments (15–41 % on three machines), while for *synctsa* it stays at 4–10 % except at four clients on one machine (20 %). Table IV therefore lists the highest throughput each configuration reached. The differences between *full*, *nobody*, *reuse* and two providers are within the spread and are not assigned to individual mechanisms. Synchronous time-stamping halves the ceiling on three machines (50 against 106 requests per second). A second provider does not raise the ceiling either: with two providers the stages of each provider become shorter, but the consumer's finishing stage becomes longer, so the consumer is the limit.

D. Environment and added round-trip time (RQ3)

With one client, three machines and one machine give the same picture (Table II): the latency of *full* differs by 0.7–1.5 ms and the differences between configurations by at most 6.4 ms. *direct* shows that the switch adds 0.6 ms at 10 KB and 9.3 ms at 1 MB, but through X-Road this is offset by the provider having a CPU of its own: on three machines the provider's stages are shorter (4.2 against 5.7 ms and 6.9 against 9.0 ms), while the way to the provider is 0.6 ms longer. Under load three machines reach 1.4 to 1.8 times higher throughput (Table IV), because the work is spread over three CPUs. With one client, all absolute values on the development laptop are lower and the CPU-bound costs smaller (logging a 1 MB body 17 instead of 35 ms, synchronous time-stamping 62 instead of 89 ms), while the differences keep their sign and order.

a) Added round-trip time.: In the three delay grids in which it was recorded, the round-trip time of the connections between the Security Servers, as smoothed by the kernel (which includes delayed acknowledgements), was 0.5, 6.1, 21.4 and 51.5 ms for *full* and 2.7, 8.1, 23.5 and 54.2 ms for *reuse* at an added 0, 5, 20 and 50 ms; in a calibration with sequential requests the minimum for new connections was 0.53, 5.54, 20.78 and 51.11 ms. With the netem rules installed and no delay, the latency did not change at 10 KB (31.0 against 30.6 ms for *full*, 27.7 against 28.9 ms for *reuse*); at 1 MB it was 3–4 ms higher (about 5 %, with overlapping ranges), one to two orders of magnitude below the effect of the added round-trip time (47 ms for *full* at 1 MB already at 5 ms).

Before the first grid with added delay, traffic captures on the local network showed that every message of *full* opens a new TCP connection and resumes the TLS 1.2 session from a session ticket, without exchanging certificates. Counting the TCP handshake, the TLS handshake and the request itself, a message of *full* waits for three round trips between the Security

Table III
PER-STAGE BREAKDOWN FROM OPERATIONAL MONITORING ON THREE MACHINES, ONE CLIENT (MEAN OF EACH RUN, MEDIAN OVER THE 15 REPETITIONS OF THE SECOND SERIES OF GRIDS, FOR WHICH THE RECORDS OF BOTH SERVERS WERE KEPT; MS; RESOLUTION 1 MS). EACH COLUMN IS A MEDIAN OF ITS OWN, SO THE COLUMNS DO NOT ADD UP EXACTLY TO THE TOTAL. "TO PROVIDER" AND "CONSUMER FINISHING" SPAN TWO MACHINES.

	Consumer sending	To provider	Provider forwarding	Echo	Provider response	Consumer finishing	Total
full, 10 KB	0.1	10.7	4.2	0.2	6.9	7.1	29.4
nobody − *full*	0.1	0.3	0.0	0.0	−0.8	−0.7	−1.2
reuse − *full*	0.0	−2.4	0.2	0.0	0.0	0.2	−2.1
synctsa − *full*	0.1	24.3	20.6	0.2	19.8	25.1	90.5
full, 1 MB	0.2	12.4	5.4	0.3	33.0	24.5	76.5
nobody − *full*	−0.1	−1.4	−1.2	−0.1	−16.9	−17.0	−37.3
reuse − *full*	0.0	−3.0	0.0	−0.1	−1.1	0.0	−3.4
synctsa − *full*	0.1	24.5	20.9	0.2	21.8	25.0	94.5

Table IV
HIGHEST MEDIAN THROUGHPUT REACHED AT 4, 16 OR 32 CLIENTS, 10 KB RESPONSE (REQUESTS PER SECOND); IN BRACKETS THE NUMBER OF CLIENTS AT WHICH IT WAS REACHED AND THE INTERQUARTILE RANGE OF THE THROUGHPUT BETWEEN REPETITIONS (%).

	full	*nobody*	*reuse*	two providers	*synctsa*
Three machines	106 (16; 32)	106 (32; 19)	108 (16; 21)	111 (32; 24)	50 (16; 10)
One machine	75 (32; 29)	76 (16; 17)	79 (16; 20)	82 (16; 36)	28 (32; 5)
Development laptop	112 (32; 9)	107 (32; 15)	118 (32; 15)	109 (32; 24)	41 (32; 5)

Servers and a message of *reuse* for one. The growth of latency is therefore described by its slope against the added round-trip time: a slope of 3 means that the latency grows by three times the added round-trip time, that is, every message waits for three round trips. It is computed from the medians at 0 and 50 ms. The prediction was a slope of 3 for *full* and 1 for *reuse*. Figure 2 and Table V show the measurement: at 10 KB the slope is 5.1 for *full* and 2.1 for *reuse*, so the prediction underestimated both. Captures under delay showed two waits it did not include: the last part of the request (842 bytes after about 10 KB) is held until the provider acknowledges the earlier data, which adds one round trip in both configurations, and on a new connection the response of about 26 KB does not fit into the initial TCP window and needs a second flight, which adds one more round trip to *full*. With these, the count is 5 and 2; a calibration with sequential requests on the three machines gave slopes of 4.7 and 1.9. A short wait inside the TLS handshake, between the message that switches the consumer to the negotiated keys and its first encrypted message, appeared at 20 ms and not at 50 ms and causes a variation of about 0.3–0.4 in the slope. The stage breakdown agrees: the slope of the way to the provider is 2.6 for *full* and 0.5 for *reuse*, and that of the provider's forwarding stage is 1 in both.

At 1 MB the slope is 10.7 for *full* and 2.0 for *reuse*. Captures of a 1 MB response at 50 ms show the reason: on a new connection the body leaves in about six flights, one per round trip, each about twice as large as the previous one (28, 57, 113, 226 and 453 KB), as the TCP window grows from its initial size; with *reuse* the window is kept between messages and keeps growing, so the response leaves in fewer flights,

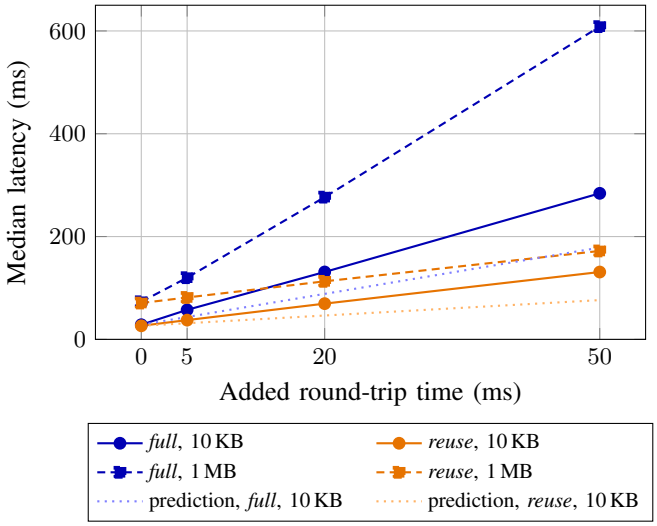


Figure 2. Latency with one client against the round-trip time added between the Security Servers, three machines, 20 repetitions; lines without marks show the prediction made before the measurement.

and from the fifth request on in one flight. The effect of the connection policy thus grows from 2 ms on the local network to 153 ms (54 %) at 10 KB and 436 ms (72 %) at 1 MB with an added 50 ms.

The extended grid confirms the pattern. With a 100 KB response the slope is 7.0 for *full* and 2.1 for *reuse*, so with a new connection for every message the cost of distance grows with the size of the response (slopes of 5.0, 7.0 and 9.7 for 10 KB, 100 KB and 1 MB in this grid), while with reused connections it does not. The slope at 1 MB is lower here than

Table V
THE POINTS OF FIGURE 2: MEDIAN LATENCY OVER 20 REPETITIONS (MS), IN BRACKETS THE FIRST AND THIRD QUARTILES OF THE RUN MEDIANS.

	0 ms	5 ms	20 ms	50 ms
full, 10 KB	28.5 (27.9–28.7)	57.4 (57.0–57.9)	131.1 (130.9–131.4)	284.1 (283.6–284.3)
reuse, 10 KB	26.4 (26.1–26.7)	37.5 (37.0–37.6)	69.5 (69.2–69.7)	131.0 (130.7–131.1)
full, 1 MB	73.1 (71.5–76.1)	120.0 (118.6–121.8)	276.7 (275.6–281.1)	608.4 (607.3–610.4)
reuse, 1 MB	70.7 (68.3–72.9)	81.5 (80.0–82.8)	112.9 (110.9–115.0)	172.0 (170.3–173.7)

in the first delay grid (9.7 against 10.7) because the medians of the two series differ by 8 ms at 0 ms and by 45 ms at 50 ms; at 10 KB the two grids agree (5.0 and 5.1). With an added round-trip time the latency with four clients is the same as with one, and at 50 ms throughput is limited by the round trips: 14 requests per second for *full* against 30 for *reuse*. With 16 clients at 0 and 20 ms the Security Servers are the limit (103–127 requests per second) and the added round-trip time disappears in the queue (142 against 140 ms for *full*); at 50 ms *full* falls to 57 requests per second while *reuse* keeps 104. Errors again occurred only with *reuse* under load: 17 of 90 000 requests at 4 and 16 clients.

With the delay applied to incoming instead of outgoing packets the slopes are the same: 5.0 for *full* and 2.1 for *reuse* at 10 KB, and 9.6 and 2.0 at 1 MB (15 repetitions, no errors). Captures of a 1 MB response at 50 ms show the same flights on a new connection (28, 57, 113, 226 and 453 KB), and with *reuse* the window is again kept between messages. The slope of *full* at 1 MB therefore does not depend on where the delay is applied: it agrees with the extended grid (9.7), while the first delay grids gave 10.7, so between nights it varies by about one round trip.

V. DISCUSSION

A. What the results mean for the proposed changes

The proposals discussed in the introduction affect different mechanisms, and the measurements put them in different positions.

Post-quantum signatures change signing and verification, which are paid on every message and run on the CPU of the Security Servers. In the measurements this work happens within the stages on the way to the provider and the provider's stages, which also contain other work, so the measurements do not give its cost alone; like all CPU-bound costs, these stages were shorter on the faster CPU of the development laptop. Larger signatures would also make every message bigger: replacing an RSA-2048 signature of 256 bytes with a Dilithium5 signature of 4595 bytes [11] on both the request and the response adds about 8.7 KB to an exchange that now carries about 40 KB in both directions. On the local network this is small. With distance between the Security Servers the size may matter more than the computation: with the default connection policy a larger response needs more flights on a new connection (Section IV-D), and the slope grew from 5 to 7 between a 10 KB and a 100 KB response. Whether a few additional kilobytes would add a flight depends on whether the message still fits into the flights it needs now, and that was not measured. The TLS handshake between the Security Servers, which post-quantum algorithms would also enlarge [10], was resumed from a session ticket without exchanging certificates in every captured message, so larger certificates would mainly affect new sessions rather than every message.

A decentralised PKI changes how certificates are checked [12]. In the present system status checking was not part of the request path: the Security Server took the status from its cache, and requests to the OCSP responder appeared only when a background refresh fell into a run. Any per-request presentation of credentials would therefore be a new cost in the request path rather than a replacement of an existing one.

Threshold signatures change signing [7]. Their reported round trips of 191 to 272 ms for small threshold groups, against 32 ms with a single software token, are several times the whole cost of X-Road measured here with one client on the local network, and of the same order as the effect of the connection policy with an added round-trip time of 50 ms.

B. The connection policy

Among the configurations compared, the largest effect is the decision to open a new connection for every message. On the local network it costs about 2 ms and 6.2 KB per exchange (3.3 KB towards the provider and 2.9 KB back), which is why it is easy to overlook. With an added round-trip time every message waits for several additional round trips, and at 50 ms reusing connections saves 54 % of the latency for a 10 KB response and 72 % for a 1 MB response, and almost doubles the throughput with 16 clients. This agrees with the observation that the cost of TLS depends mostly on whether connections are reused [14]. Connection reuse is off by default [13], so its benefit grows with the distance between the Security Servers of the communicating organisations. It is not free of side effects either: in the measurements, single requests over reused connections failed, consistent with a pooled connection that had already been closed by the other side.

The connection policy also decides how much larger post-quantum messages would cost. With reused connections a few additional kilobytes leave in the same flight, while with a new connection for every message and distance between the Security Servers they could cost additional round trips on every message if the messages no longer fit into the same flights (Section V-A).

C. Comparison with earlier measurements

The end-to-end latency with one client, 25 to 30 ms in the three environments used here, is of the same order as the 32 ms reported for X-Road 7.7.0 with all components on one laptop [7]. That figure was presented as a lower bound because there was no network between the servers [7]. For one client on a local network, the comparison of one machine with three machines shows that such a single-host figure is representative, since the network cost is offset by the provider having a CPU of its own. It is not representative of throughput, which was 1.4 to 1.8 times higher on three machines, nor of Security Servers that are farther apart, where the connection policy dominates. In the earlier measurement of a Security Server in front of a registry service, the path through the Security Server was reported as faster than the direct one at 20 and 30 concurrent clients [8], which suggests that the variation of the environment was comparable to the effect being measured; here, too, the spread under load was too large to compare configurations, while with one client and with an added round-trip time it mostly stayed within a few per cent.

The warm-up needed was much longer than the earlier work suggests. The closest measurement reports means over 1000 requests without describing a warm-up [7], while here the latency did not settle within the first thousand requests and the first run after a restart stayed slower even after 4000, in line with the advice to measure Java systems only in the steady state [20]. Short measurements of X-Road may therefore overestimate its latency and vary more between runs.

D. Absolute values and differences

The absolute values depend on the hardware: the same grid gave 24.8 ms on the development laptop and 28.3 ms on the three machines for a 10 KB response, and the CPU-bound costs, logging a large body and synchronous time-stamping, differed by up to a factor of two. The differences between configurations kept their sign and order in every environment, and the effect of the connection policy on the local network was about the same (1.6–2.0 ms at 10 KB). The power mode of the CPU had an even larger effect than the choice of machine. Measurements of X-Road on general-purpose hardware should therefore report the CPU and its power mode together with the results.

VI. Conclusion

This study measured how much the trust mechanisms of X-Road cost per message, on three laptops connected by a switch, on one of them alone, on a faster laptop, and with a controlled round-trip time added between the Security Servers.

For RQ1, the X-Road path added about 28 ms to a 10 KB exchange with one client on these machines. Logging the message body made no measurable difference for small messages and accounted for almost half of the latency of a 1 MB exchange. Opening a new connection for every message cost about 2 ms and 6.2 KB per exchange on the local network. Certificate status checking and batched time-stamping were not in the request path: requests to the OCSP responder only from background refreshes and about two time-stamping requests per 1200 messages, while synchronous time-stamping added about 90 ms. Signing and verification cannot be switched off; the stages that contain them take 4 to 11 ms each, which bounds their cost but does not measure the operations alone.

For RQ2, most of the growth from a 10 KB to a 1 MB response came from writing the body to the message log. Under load the Security Servers reached about 100 requests per second on three machines, synchronous time-stamping halved that, and the spread between repetitions was too large to compare the other configurations.

For RQ3, one machine and three machines gave nearly the same latency and differences with one client, while three machines reached a higher throughput. With an added round-trip time the default connection policy made every message wait for about five round trips at 10 KB and ten at 1 MB, against two with reused connections; a prediction from the handshakes alone gave three and one and missed two waits inside TCP.

The results give a baseline for the present system against which post-quantum signatures, a decentralised PKI and threshold signatures can be compared. They also show that the cost of larger post-quantum messages will depend on the connection policy: small with reused connections, but possibly additional round trips on every message when each message opens a new connection, the Security Servers are far apart and the larger messages no longer fit into the same flights. Further work could measure post-quantum signatures once they are available in X-Road, repeat the measurements with external certification and time-stamping services and over a real wide-area link, and use server hardware.

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